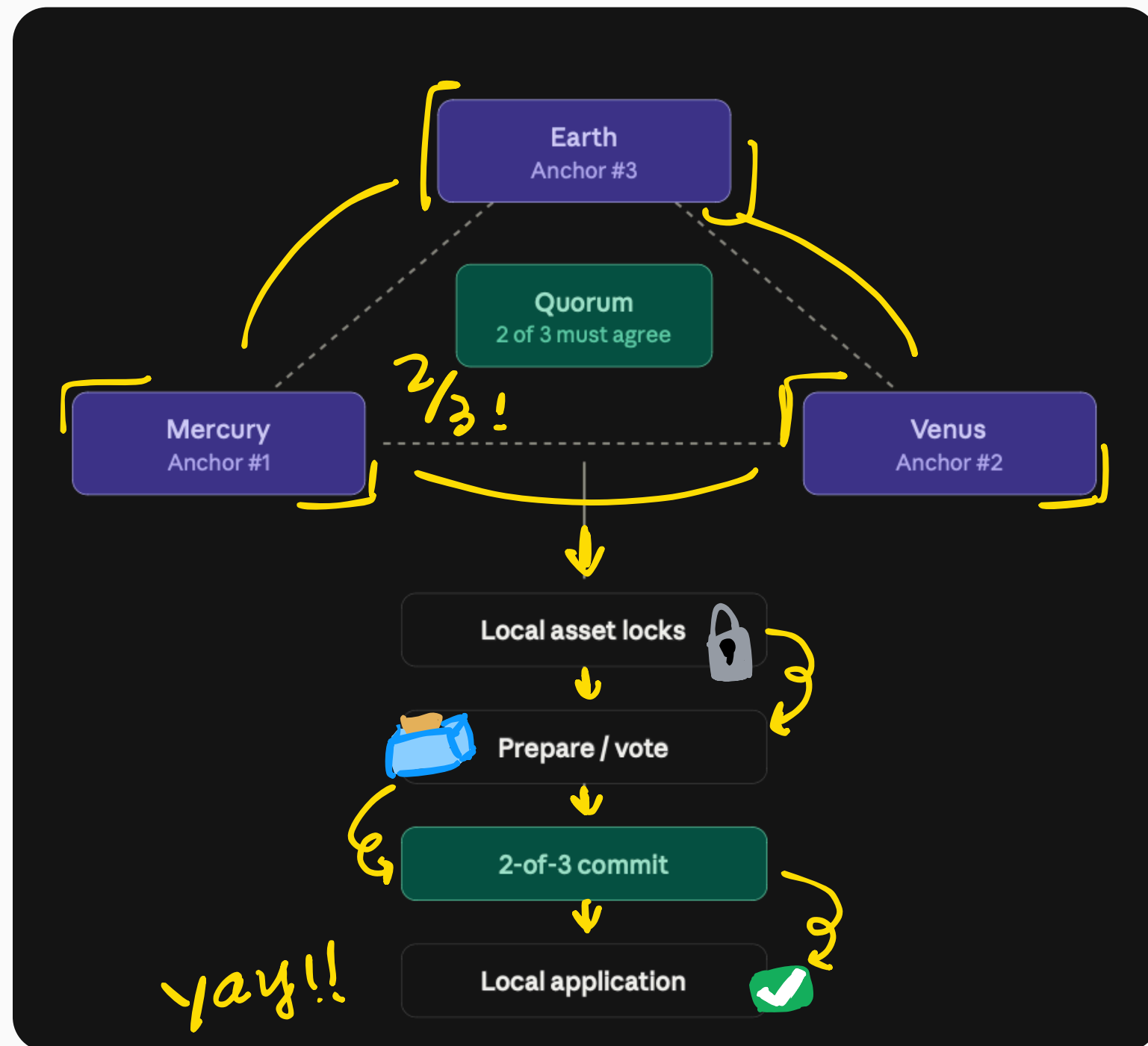


A “Brief” Overview of Iron Triangle

WE TRIED OUR BEST 🙄



The Iron Triangle of Doom and Despair

The most distinguishing factor of our design choice was to ensure that even if packets can be lost, a route can disappear, or a planet can go offline, none of those events can create a blackout where transactions cannot occur.

Dependable ↑
max

Why these 3 in particular?

We simulated 2000 years of position data

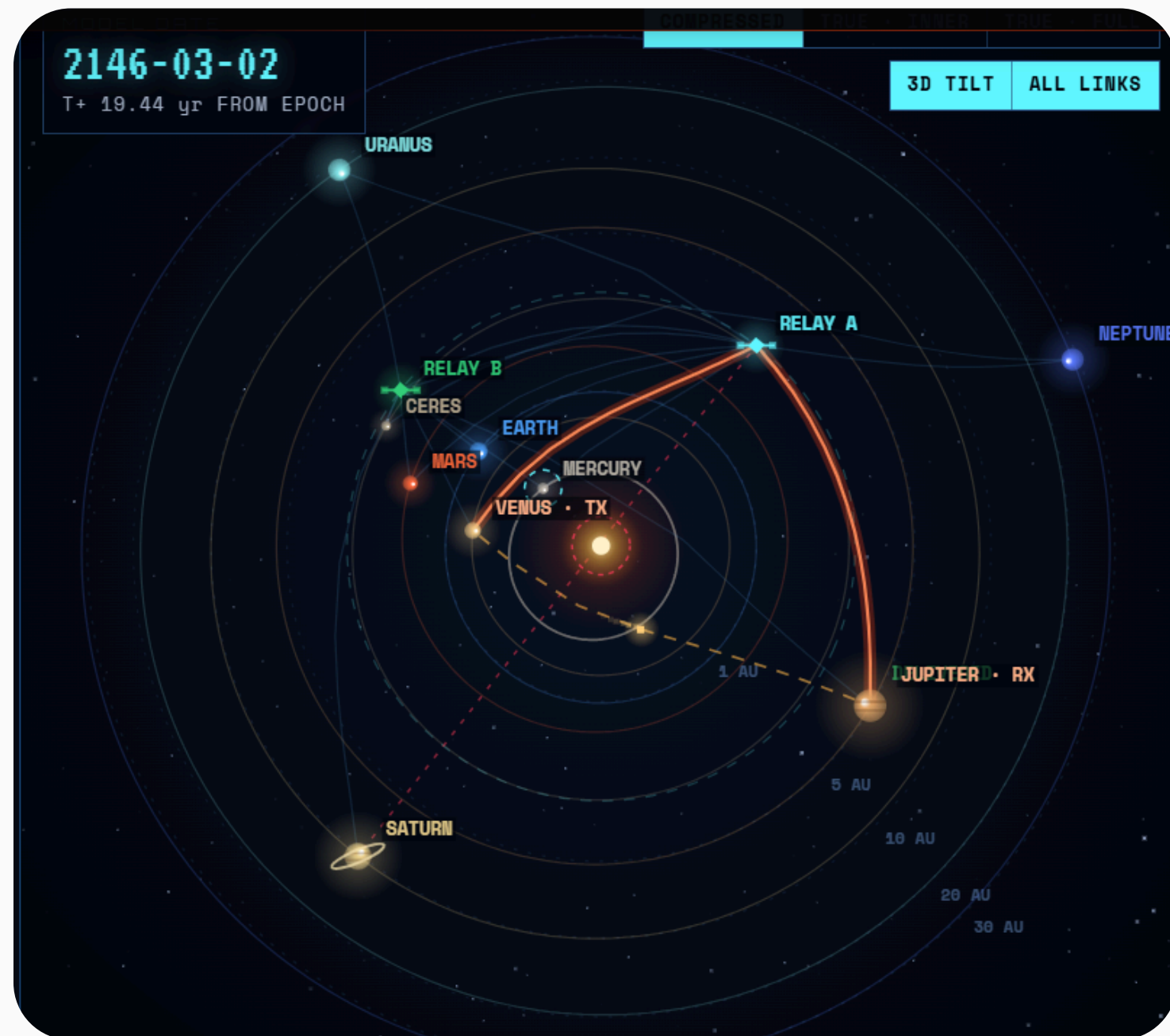
20000
2000?!

Sec1	Dist1	Pair	Delay_L	Delay_M	Delay_S	Delay_H	Direct_Lurr	Direct_Lurr	Direct_Lurr_F	Direct_Lurr_H	Backbone_Lurr	Backbone_Lurr	Backbone_Lurr	Backbone_Lurr_H	Total_At	Failed_At	Blackout_L	MultiHop_R	Barista_Pct
Ceres	Earth	Ceres-Earth	14.71	36.62	23.82	24.48	13.18%	28.21%	20.30%	20.93%	7.49%	8.42%	7.64%	7.62%	730000	0	0.00%	31667	4.34%
Ceres	Jupiter	Ceres-Jupiter	20.28	68.41	46.44	49.07	17.71%	48.20%	35.21%	37.54%	14.73%	36.24%	14.80%	14.76%	730000	0	0.00%	12882	1.76%
Ceres	Mars	Ceres-Mars	10.36	38.51	24.82	26.27	9.47%	30.88%	20.89%	22.24%	8.21%	20.65%	8.29%	8.29%	730000	0	0.00%	23645	3.24%
Ceres	Mercury	Ceres-Mercury	19.81	36.93	23.41	23.25	17.34%	29.52%	19.98%	19.90%	6.09%	16.89%	6.68%	6.58%	730000	0	0.00%	69115	9.47%
Ceres	Neptune	Ceres-Neptune	227.08	274.4	250.63	251.15	88.74%	92.86%	90.91%	91.06%	48.14%	91.09%	62.71%	48.15%	730000	0	0.00%	9172	1.26%
Ceres	Saturn	Ceres-Saturn	56.32	103.61	81.03	82.64	41.82%	63.08%	53.54%	54.78%	21.81%	54.92%	22.00%	21.83%	730000	0	0.00%	10856	1.49%
Ceres	Uranus	Ceres-Uranus	136.61	183.88	160.44	161.24	73.12%	82.94%	78.35%	78.77%	35.54%	79.16%	35.56%	35.56%	730000	0	0.00%	9596	1.31%
Ceres	Venus	Ceres-Venus	17.01	35.93	23.48	23.8	15.08%	29.20%	20.07%	20.37%	6.73%	20.34%	7.17%	6.95%	730000	0	0.00%	40622	5.56%
Earth	Ceres	Earth-Ceres	14.71	36.62	23.82	24.48	13.18%	28.21%	20.30%	20.93%	7.49%	8.42%	7.64%	7.62%	730000	0	0.00%	31667	4.34%
Earth	Jupiter	Earth-Jupiter	24.97	56.24	43.71	44.08	28.56%	40.84%	34.19%	34.51%	11.83%	12.64%	11.94%	11.92%	730000	0	0.00%	27739	3.80%
Earth	Mars	Earth-Mars	4.37	26.61	14.15	15.17	4.10%	21.49%	12.55%	13.53%	5.33%	7.95%	5.54%	5.51%	730000	0	0.00%	38537	5.28%
Earth	Mercury	Earth-Mercury	5.11	22.92	9.22	8.93	4.78%	18.39%	8.04%	8.00%	3.93%	7.09%	4.09%	4.09%	730000	0	0.00%	83990	11.51%
Earth	Neptune	Earth-Neptune	241.78	262.25	250.18	250.23	90.23%	91.90%	90.97%	90.99%	46.35%	46.55%	60.33%	46.39%	730000	0	0.00%	24056	3.30%
Earth	Saturn	Earth-Saturn	71.02	91.98	79.57	79.77	49.49%	57.96%	53.39%	53.54%	19.13%	19.84%	19.21%	19.20%	730000	0	0.00%	25717	3.52%
Earth	Uranus	Earth-Uranus	151.31	172.2	159.75	159.84	76.67%	80.59%	78.44%	78.49%	33.32%	35.91%	43.41%	33.37%	730000	0	0.00%	24476	3.35%
Earth	Venus	Earth-Venus	2.32	22.51	9.62	10.28	2.19%	14.62%	8.59%	9.28%	4.49%	5.26%	4.66%	4.66%	730000	0	0.00%	55493	7.60%
Jupiter	Ceres	Jupiter-Ceres	20.28	68.41	46.44	49.07	17.71%	48.20%	35.21%	37.54%	14.73%	36.24%	14.80%	14.76%	730000	0	0.00%	12882	1.76%
Jupiter	Earth	Jupiter-Earth	24.97	56.24	43.71	44.08	28.56%	40.84%	34.19%	34.51%	11.83%	12.64%	11.94%	11.92%	730000	0	0.00%	27739	3.80%
Jupiter	Mars	Jupiter-Mars	30.62	59.14	44.23	45.11	25.50%	42.94%	34.36%	35.13%	12.58%	34.69%	12.78%	12.64%	730000	0	0.00%	19723	2.70%
Jupiter	Mercury	Jupiter-Mercur	40.07	57.15	43.57	43.41	31.98%	42.27%	34.10%	34.02%	10.55%	14.05%	11.05%	10.94%	730000	0	0.00%	65184	8.93%
Jupiter	Neptune	Jupiter-Neptun	206.82	293.97	252.05	253.94	86.32%	94.08%	90.75%	91.30%	50.61%	91.51%	65.85%	50.62%	730000	0	0.00%	5181	0.71%
Jupiter	Saturn	Jupiter-Saturn	36.06	123.49	85.48	90.57	29.30%	69.50%	54.16%	58.10%	25.53%	57.04%	25.61%	25.54%	730000	0	0.00%	6927	0.95%
Jupiter	Uranus	Jupiter-Uranus	116.35	203.58	162.5	165.25	67.34%	85.88%	78.12%	79.57%	38.61%	65.60%	50.23%	38.62%	730000	0	0.00%	5648	0.77%
Jupiter	Venus	Jupiter-Venus	37.28	56.37	43.55	43.71	30.12%	41.44%	34.11%	34.25%	11.16%	34.51%	11.56%	11.32%	730000	0	0.00%	36694	5.03%
Mars	Ceres	Mars-Ceres	10.36	38.51	24.82	26.27	9.47%	30.88%	20.89%	22.24%	8.21%	20.65%	8.29%	8.29%	730000	0	0.00%	23645	3.24%
Mars	Earth	Mars-Earth	4.37	26.61	14.15	15.17	4.10%	21.49%	12.55%	13.53%	5.33%	6.19%	5.54%	5.51%	730000	0	0.00%	38537	5.28%
Mars	Jupiter	Mars-Jupiter	30.62	59.14	44.23	45.11	25.50%	42.94%	34.36%	35.13%	12.58%	34.69%	12.78%	12.64%	730000	0	0.00%	19723	2.70%
Mars	Mercury	Mars-Mercury	9.47	26.92	13.28	13.09	8.69%	22.61%	11.75%	11.67%	3.72%	8.40%	4.57%	4.47%	730000	0	0.00%	75980	10.41%
Mars	Neptune	Mars-Neptune	237.42	265.14	250.26	250.42	89.81%	92.18%	90.95%	91.00%	46.84%	91.06%	61.36%	46.86%	730000	0	0.00%	16044	2.20%
Mars	Saturn	Mars-Saturn	66.66	94.87	79.85	80.34	47.33%	59.84%	53.40%	53.78%	19.84%	53.81%	20.08%	19.88%	730000	0	0.00%	17702	2.42%
Mars	Uranus	Mars-Uranus	146.95	175.15	159.88	160.12	75.67%	81.45%	78.42%	78.55%	33.92%	78.66%	44.25%	33.95%	730000	0	0.00%	16456	2.25%
Mars	Venus	Mars-Venus	6.67	26.42	13.51	14.04	6.20%	22.41%	12.01%	12.53%	4.38%	12.29%	5.10%	4.97%	730000	0	0.00%	47483	6.50%
Mercury	Ceres	Mercury-Ceres	19.81	36.93	23.41	23.25	17.34%	29.52%	19.98%	19.90%	6.09%	16.89%	6.68%	6.58%	730000	0	0.00%	69115	9.47%
Mercury	Earth	Mercury-Earth	5.11	22.92	9.22	8.93	4.78%	18.39%	8.04%	8.00%	3.93%	5.36%	4.09%	4.09%	730000	0	0.00%	83990	11.51%
Mercury	Jupiter	Mercury-Jupite	40.07	57.15	43.57	43.41	31.98%	42.27%	34.10%	34.02%	10.55%	14.05%	11.05%	10.94%	730000	0	0.00%	65184	8.93%
Mercury	Mars	Mercury-Mars	9.47	26.92	13.28	13.09	8.69%	22.61%	11.75%	11.67%	3.72%	8.40%	4.57%	4.47%	730000	0	0.00%	75980	10.41%
Mercury	Neptune	Mercury-Nepts	206.82	293.97	252.05	253.94	86.32%	94.08%	90.75%	91.30%	50.61%	91.51%	65.85%	50.62%	730000	0	0.00%	5181	0.71%
Mercury	Saturn	Mercury-Satur	36.06	123.49	85.48	90.57	29.30%	69.50%	54.16%	58.10%	25.53%	57.04%	25.61%	25.54%	730000	0	0.00%	6927	0.95%
Mercury	Uranus	Mercury-Urans	116.35	203.58	162.5	165.25	67.34%	85.88%	78.12%	79.57%	38.61%	65.60%	50.23%	38.62%	730000	0	0.00%	5648	0.77%
Mercury	Venus	Mercury-Venus	6.67	26.42	13.51	14.04	6.20%	22.41%	12.01%	12.53%	4.38%	12.29%	5.10%	4.97%	730000	0	0.00%	47483	6.50%
Neptune	Ceres	Neptune-Ceres	227.08	274.4	250.63	251.15	88.74%	92.86%	90.91%	91.06%	48.14%	91.09%	62.71%	48.15%	730000	0	0.00%	9172	1.26%
Neptune	Earth	Neptune-Earth	241.78	262.25	250.18	250.23	90.23%	91.90%	90.97%	90.99%	46.35%	46.77%	46.40%	46.39%	730000	0	0.00%	24056	3.30%
Neptune	Jupiter	Neptune-Jupite	206.82	293.97	252.05	253.94	86.32%	94.08%	90.75%	91.30%	50.61%	91.51%	65.85%	50.62%	730000	0	0.00%	5181	0.71%
Neptune	Mars	Neptune-Mars	237.42	265.14	250.26	250.42	89.81%	92.18%	90.95%	91.00%	46.84%	91.06%	61.36%	46.86%	730000	0	0.00%	16044	2.20%
Neptune	Mercury	Neptune-Mercur	246.87	263.52	250.27	250.11	90.69%	92.05%	90.98%	90.97%	45.60%	88.15%	45.86%	45.79%	730000	0	0.00%	61486	8.42%
Neptune	Saturn	Neptune-Satur	170.78	329.79	256.81	263.03	80.65%	95.81%	90.25%	92.03%	54.71%	54.74%	54.72%	54.71%	730000	0	0.00%	3228	0.44%
Neptune	Uranus	Neptune-Urans	90.48	409.99	272.86	290.45	58.11%	98.06%	87.74%	93.87%	62.66%	62.68%	81.47%	62.66%	730000	0	0.00%	1873	0.26%
Neptune	Venus	Neptune-Venus	244.08	262.54	250.16	250.17	90.44%	91.96%	90.97%	90.98%	45.97%	91.03%	60.00%	46.04%	730000	0	0.00%	33006	4.52%
Saturn	Ceres	Saturn-Ceres	56.32	103.61	81.03	82.64	41.82%	63.08%	53.54%	54.78%	21.81%	54.92%	22.00%	21.83%	730000	0	0.00%	10856	1.49%
Saturn	Earth	Saturn-Earth	71.02	91.98	79.57	79.77	49.49%	57.96%	53.39%	53.54%	19.13%	19.84%	19.21%	19.20%	730000	0	0.00%	25717	3.52%
Saturn	Jupiter	Saturn-Jupiter	36.06	123.49	85.48	90.57	29.30%	69.50%	54.16%	58.10%	25.53%	57.04%	25.61%	25.54%	730000	0	0.00%	6927	0.95%
Saturn	Mars	Saturn-Mars	66.66	94.87	79.85	80.34	47.33%	59.84%	53.40%	53.78%	19.84%	53.81%	20.08%	19.88%	730000	0	0.00%	17702	2.42%
Saturn	Mercury	Saturn-Mercur	76.11	92.53	79.56	79.4	51.91%	58.92%	53.39%	53.33%	17.98%	51.58%	18.40%	18.30%	730000	0	0.00%	63148	8.65%
Saturn	Neptune	Saturn-Neptun	170.78	329.79	256.81	263.03	80.65%	95.81%	90.25%	92.03%	54.71%	54.74%	71.14%	54.71%	730000	0	0.00%	3228	0.44%
Saturn	Uranus	Saturn-Uranus	80.31	239.3	169.7	178.35	53.81%	89.99%	77.49%	81.99%	43.70%	81.42%	56.97%	43.71%	730000	0	0.00%	3726	0.51%
Saturn	Venus	Saturn-Venus	73.32	91.65	79.49	79.56	50.59%	58.58%	53.37%	53.43%	18.54%	53.65%	18.82%	18.66%	730000	0	0.00%	34669	4.75%
Uranus	Ceres	Uranus-Ceres	136.61	183.88	160.44	161.24	73.12%	82.94%	78.35%	78.77%	35.54%	79.16%	35.74%	35.56%	730000	0	0.00%	9596	1.31%
Uranus	Earth	Uranus-Earth	151.31	172.2	159.75	159.84	76.67%	80.61%	78.44%	78.49%	33.32%	33.90%	33.37%	33.37%	730000	0	0.00%	24476	3.35%
Uranus	Jupiter	Uranus-Jupiter	116.35	203.58	162.5	165.25	67.34%	85.88%	78.12%	79.57%	38.61%	65.60%	38.62%	38.62%	730000	0	0.00%	5648	0.77%
Uranus	Mars	Uranus-Mars	146.95	175.15	159.88	160.12	75.67%	81.45%	78.42%	78.55%	33.92%	78.66%	34.05%	33.95%	730000	0	0.00%	16456	2.25%
Uranus	Mercury	Uranus-Mercur	156.41	173.37	159.81	159.66	77.78%	81.11%	78.46%	78.44%	32.38%	76.87%	32.71%	32.62%	730000	0	0.00%	61913	8.48%
Uranus	Neptune	Uranus-Neptun	90.48	409.99	272.86	290.45	58.11%	98.06%	87.74%	93.87%	62.66%	62.68%	81.47%	62.66%	730000	0	0.00%	1873	0.26%
Uranus	Saturn	Uranus-Saturn	80.31	239.3	169.7	178.35	53.81%	89.99%	77.49%	81.99%	43.70%	81.42%	43.71%	43.71%	730000	0	0.00%	3726	0.51%
Uranus	Venus	Uranus-Venus	153.61	172.32	159.72	159.74	77.18%	80.93%	78.45%	78.47%	32.84%	78.55%	33.06%	32.93%	730000				

Source ▾	Destination ▾	Pair ▾	Delay_Mean ▾	Direct_Loss_Mean_Pc ▾	Backbone_Loss_Mean_Pc ▾
Ceres	Earth	Ceres-Earth	23.82	20.30%	7.64%
Ceres	Jupiter	Ceres-Jupiter	46.44	35.21%	14.80%
Ceres	Mars	Ceres-Mars	24.82	20.89%	8.39%
Ceres	Mercury	Ceres-Mercury	23.41	19.98%	6.68%
Ceres	Neptune	Ceres-Neptune	250.63	90.91%	62.71%
Ceres	Saturn	Ceres-Saturn	81.03	53.54%	22.00%
Ceres	Uranus	Ceres-Uranus	160.44	78.35%	46.46%
Ceres	Venus	Ceres-Venus	23.48	20.07%	7.17%
Earth	Ceres	Earth-Ceres	23.82	20.30%	7.64%
Earth	Jupiter	Earth-Jupiter	43.71	34.19%	11.94%
Earth	Mars	Earth-Mars	14.15	12.55%	5.54%
Earth	Mercury	Earth-Mercury	9.22	8.04%	4.09%
Earth	Neptune	Earth-Neptune	250.18	90.97%	60.33%
Earth	Saturn	Earth-Saturn	79.57	53.39%	19.21%
Earth	Uranus	Earth-Uranus	159.75	78.44%	43.41%
Earth	Venus	Earth-Venus	9.62	8.59%	4.66%
Jupiter	Ceres	Jupiter-Ceres	46.44	35.21%	14.80%
Jupiter	Earth	Jupiter-Earth	43.71	34.19%	11.94%
Jupiter	Mars	Jupiter-Mars	44.23	34.36%	12.79%
Jupiter	Mercury	Jupiter-Mercury	43.57	34.10%	11.05%
Jupiter	Neptune	Jupiter-Neptune	252.05	90.75%	65.85%
Jupiter	Saturn	Jupiter-Saturn	85.48	54.16%	25.61%
Jupiter	Uranus	Jupiter-Uranus	162.5	78.12%	50.23%
Jupiter	Venus	Jupiter-Venus	43.55	34.11%	11.56%
Mars	Ceres	Mars-Ceres	24.82	20.89%	8.39%

Planet ▾	MeanBackBoneLoss ▾	MaxBackBoneLoss ▾
Mercury	0.18870125	0.1889125
Venus	0.1928425	0.3292625
Earth	0.196025	0.182075
Mars	0.20259	0.3821875
Ceres	0.2198175	0.4096375
Jupiter	0.25478375	0.43285
Saturn	0.3152775	0.53375
Uranus	0.41610125	0.69605
Neptune	0.52586375	0.7712875





Being blind in space, not a good combo x

What happens when Earth and Mercury can't see the outer system?

Our system automatically switches to Venus.

- The main takeaway is that the architecture doesn't need us to predict which planet will be best.
- The same path-election rule automatically chooses the surviving route



Trade	Earth	Mercury	Venus	Winner
Earth-Venus	2.028 h	3.215 h	2.308 h	Earth
Mercury-Venus	1.380 h	0.086 h	0.086 h	Mercury
Earth-Mars	1.130 h	2.609 h	2.426 h	Earth
Mars-Neptune	9.491 h	8.496 h	8.323 h	Venus

The initiator moves with the trade:

- For every trade the originating settlement calculates

$$T(A) = d(S_1, A) + 2 \min_{B \neq A} d(A, B) + d(A, S_2)$$

- For a Mercury-Venus trade, an Earth-only architecture takes 1.38 hours.
- Our rotating architecture takes 0.086 hours, a savings of about 1.3 hours.

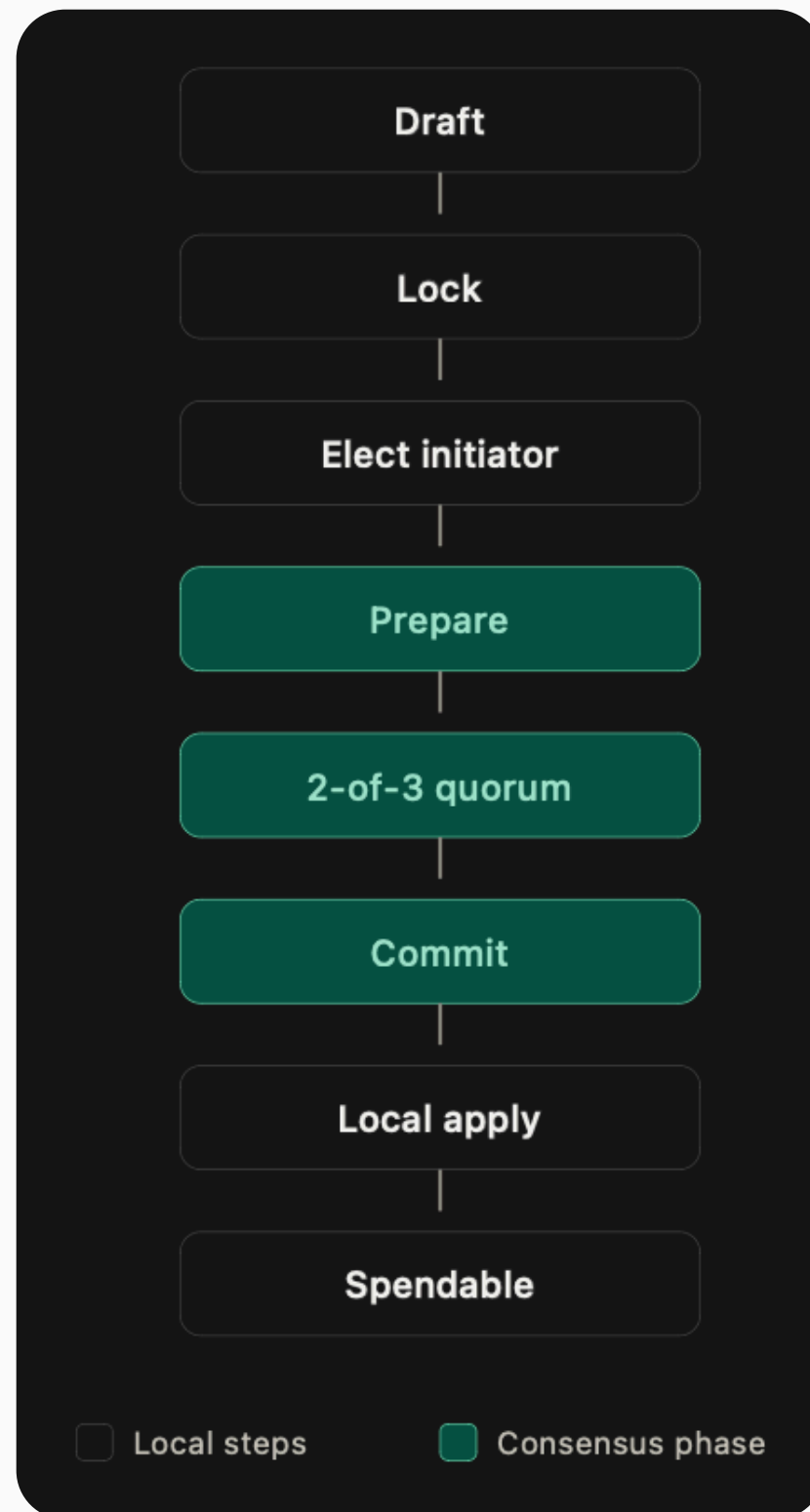
What Makes our Architecture so Amazing ✨ ✨ ✨ ✨ ✨ 😜

Less waiting
Shorter distance
Lower packet loss chances

$$y = 1 - e^{-0.02x}$$

$$y = 1 - e^{-0.08x}$$





Quantum money sounds cool, but we don't do that

Guarantees:

1. No double spending ✗
2. No premature settlement ✗
3. No phantom collateral ✗

A packet can be lost, a route can disappear, or a planet can go offline. However, none of those events can create or destroy money.



Risk	Our response	Residual cost
Gateway isolation	2-of-3 quorum	Availability for affected settlement
Solar occultation	Alternate anchors/relays	Longer settlement time
Packet loss	Retries + duplicate-safe IDs	More traffic
Long latency	Durable locks	Capital trapped longer
Margin shortfall	Pre-trade rejection	Trade may not open
Default	Funded waterfall	Recovery may be delayed
Capacity exhaustion	Queue/reject before locking	Reduced throughput

It’s just a game of give and take

Our most important tradeoff: We deliberately prioritize correctness over immediate availability. Funds remain safely locked and settlement waits if communication disappears.

Our promises to you! ✨ ✨ ✨

✧ Our design is crafted to make financial state trustworthy when space is slow. 🐢
We tested that claim with orbital propagation, packet-loss, probabilities, long-horizon availability, stress scenarios, and balance-sheet conservation.

1,753,2010

hours simulated in the spreadsheet
using c++

99.51%

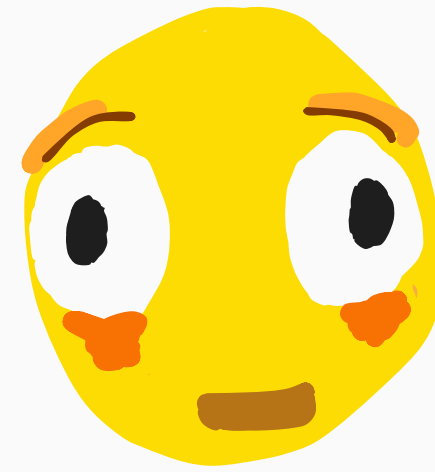
of backbone sent packets are received

1.1 h

recovery after isolation

0%

Backbone blackout



THE BESTEST OF STATS

THE COOLEST THING YOU'LL
SEE TODAY

Thank you for your time